

Lecture 4: Constrained Optimization I

Junnan Zhang

Paula and Gregory Chow Institute for Studies in Economics
Xiamen University

Slides Prepared by Xiaoling Mei

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- 1 Definition
- 2 Equality Constraints
- 3 Inequality Constraints
- 4 Mixed Constraints

We often work with the optimization problem in which the objects are not free to take on any value but are constrained. For example:

- ▶ A household's consumption is constrained by available income
- ▶ A firm's production is constrained by the cost and availability of its inputs
- ▶ The central mathematical problem here is that of maximizing/minimizing a function of several variables, where these variables are bound by some constraining equations.

The Prototype Problem

- ▶ Objective function: $f(x_1, x_2, \dots, x_n)$
- ▶ Constraint functions:

$$g_1(x_1, \dots, x_n) \leq b_1$$

$$\vdots$$

$$g_k(x_1, \dots, x_n) \leq b_k$$

$$h_1(x_1, \dots, x_n) = c_1$$

$$\vdots$$

$$h_m(x_1, \dots, x_n) = c_m$$

- 1 Definition
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- 3 Inequality Constraints
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- ▶ Simplest constrained maximization problem: maximizing a function of two variables subject to a single equality constraint
- ▶ Setup:

$$\begin{array}{ll}\max & f(x_1, x_2) \\ \text{s.t.} & h(x_1, x_2) = c\end{array}$$

- ▶ The highest level set of f must touch (be tangent to) the constraint curve C at the constrained max. Figure 18.1.

- ▶ Equivalently, at the constrained max $\mathbf{x}^*$, the slope of the level set of f equals the slope of the constraint curve C .
- ▶ If $\frac{\partial f}{\partial x_2}(\mathbf{x}^*) \neq 0$, the slope of the level set of f at $\mathbf{x}^*$ is

$$-\frac{\partial f}{\partial x_1}(\mathbf{x}^*) / \frac{\partial f}{\partial x_2}(\mathbf{x}^*)$$

- ▶ If $\frac{\partial h}{\partial x_2}(\mathbf{x}^*) \neq 0$, the slope of the constraint set $h(x_1, x_2) = c$ at $\mathbf{x}^*$ is

$$-\frac{\partial h}{\partial x_1}(\mathbf{x}^*) / \frac{\partial h}{\partial x_2}(\mathbf{x}^*)$$

- Equalize the slope and we get:

$$\frac{\frac{\partial f}{\partial x_1}(\mathbf{x}^*)}{\frac{\partial h}{\partial x_1}(\mathbf{x}^*)} = \frac{\frac{\partial f}{\partial x_2}(\mathbf{x}^*)}{\frac{\partial h}{\partial x_2}(\mathbf{x}^*)} = \mu.$$

- Rewrite the equation and combine with the constraint equation:

$$\frac{\partial f}{\partial x_1}(\mathbf{x}^*) - \mu \frac{\partial h}{\partial x_1}(\mathbf{x}^*) = 0$$

$$\frac{\partial f}{\partial x_2}(\mathbf{x}^*) - \mu \frac{\partial h}{\partial x_2}(\mathbf{x}^*) = 0$$

$$h(x_1, x_2) = c$$

A system with three unknowns.

- More generally, tangency means that the gradients are proportional:

$$\nabla f(\mathbf{x}^*) = \mu \nabla h(\mathbf{x}^*).$$

Constraint Qualification

- ▶ Above method would not have worked if both $\partial h / \partial x_1$ and $\partial h / \partial x_2$ were 0;
- ▶ We will need to make the assumption that $\partial h / \partial x_1$ or $\partial h / \partial x_2$ (or both) is not zero at the maximizer
- ▶ This restriction is called a **constraint qualification**

Theorem 18.1

Suppose $\mathbf{x}^* = (x_1^*, x_2^*)$ is a solution and suppose further that it is not a critical point of h . Then there is real number μ^* such that (x_1^*, x_2^*, μ^*) is a critical point of the *Lagrangian function*

$$L(x_1, x_2, \mu) \equiv f(x_1, x_2) - \mu(h(x_1, x_2) - c).$$

In other words, at (x_1^*, x_2^*, μ^*) :

$$\frac{\partial L}{\partial x_1} = 0, \frac{\partial L}{\partial x_2} = 0, \frac{\partial L}{\partial \mu} = 0$$

μ is called a **Lagrange multiplier**.

Example 18.4: Use Theorem 18.1 to solve a simple utility maximization problem:

$$\begin{array}{ll}\text{maximize} & f(x_1, x_2) = x_1 x_2 \\ \text{subject to} & h(x_1, x_2) \equiv x_1 + 4x_2 = 16\end{array}$$

The gradient of h is $(1, 4)$, so h has no critical points and the constraint qualification is satisfied. From the Lagrangian function

$$L(x_1, x_2, \mu) = x_1 x_2 - \mu(x_1 + 4x_2 - 16)$$

Example 18.4 : Set the partial derivatives equal to zero:

$$\frac{\partial L}{\partial x_1} = x_2 - \mu = 0$$

$$\frac{\partial L}{\partial x_2} = x_1 - 4\mu = 0$$

$$\frac{\partial L}{\partial \mu} = -(x_1 + 4x_2 - 16) = 0$$

We conclude the solution of this system is

$$x_1 = 8, x_2 = 2, \mu = 2$$

So the only candidate for a solution is

$$x_1 = 8, x_2 = 2$$

- ▶ When there are several equality constraints, the problem becomes

$$\begin{aligned} & \max f(x_1, x_2, \dots, x_n) \\ & \text{s.t. } h_1(\mathbf{x}) = a_1; \dots; h_m(\mathbf{x}) = a_m \end{aligned}$$

- ▶ Generalized constraint qualification: x is called a critical point of $\mathbf{h} = (h_1, h_2, \dots, h_m)$ if the rank of the matrix $D\mathbf{h}(\mathbf{x}^*)$ is less than m .
- ▶ **Nondegenerate Constraint Qualification:** The rank of Jacobian $D\mathbf{h}(\mathbf{x}^*)$ is equal to the number of the constraints, then $\mathbf{x}^*$ satisfies NDCQ.

Theorem 18.2

Suppose x^* is a solution and suppose further it satisfies condition NDCQ. Then there exists $\mu^* = (\mu_1^*, \dots, \mu_m^*)$ such that (x^*, μ^*) is a critical point of the *Lagrangian*

$$L(x, \mu) \equiv f(x) - \mu_1(h_1(x) - a_1) - \mu_2(h_2(x) - a_2) - \dots - \mu_m(h_m(x) - a_m).$$

In other words,

$$\frac{\partial L}{\partial x_1}(x^*, \mu^*) = 0, \dots, \frac{\partial L}{\partial x_n}(x^*, \mu^*) = 0$$

$$\frac{\partial L}{\partial \mu_1}(x^*, \mu^*) = 0, \dots, \frac{\partial L}{\partial \mu_m}(x^*, \mu^*) = 0$$

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- ▶ Many optimization problems have their constraints defined by inequalities
 $g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k$
- ▶ The method for finding the constrained maxima is more complex
- ▶ We start with the simplest case when there are two variables and one *inequality* constraint

- Consider the optimization problem with one inequality constraint:

$$\begin{aligned} \max \quad & f(x, y) \\ \text{s.t.} \quad & g(x, y) \leq b \end{aligned}$$

Case A: If the maximum of f occurs at a point where $g(x, y) = b$, that is, when the constraint is **binding** (active/effective/tight)

- ▶ Highest level curve of f meets the constraint set at point $\mathbf{p}$
- ▶ Equivalently, the level set of f and the level set of g are tangent to each other at $\mathbf{p}$
- ▶ $\nabla f(\mathbf{p})$, $\nabla g(\mathbf{p})$ line up: $\nabla f(\mathbf{p}) - \lambda \nabla g(\mathbf{p}) = \mathbf{0}$
- ▶ More importantly, $\nabla f(\mathbf{p})$ and $\nabla g(\mathbf{p})$ *point to the same direction*: $\lambda \geq 0$
- ▶ Form the Lagrangian function:

$$L(x, y, \lambda) = f(x, y) - \lambda[g(x, y) - b]$$

Case B: If the maximum of f occurs at a point where $g(x, y) < b$, that is, when the constraint is **not binding** (inactive/ineffective/loose)

- ▶ $\mathbf{q}$ must be a local max of f , that is, a local unconstrained max
- ▶ Then we have $\frac{\partial f}{\partial x}(\mathbf{q}) = 0$ and $\frac{\partial f}{\partial y}(\mathbf{q}) = 0$
- ▶ The derivative of g does not enter the calculations at $\mathbf{q}$
- ▶ In this case, we can still use the Lagrangian function:

$$L(x, y, \lambda) = f(x, y) - \lambda[g(x, y) - b]$$

provided that we set $\lambda = 0$.

In summary:

- ▶ The constraint is binding: $g(x, y) - b = 0$, in this case $\lambda \geq 0$
- ▶ The constraint is not binding, in this case $\lambda = 0$
- ▶ Therefore $\lambda = 0$ or $g(x, y) - b = 0$ (and both may hold). This is called a **complementary slackness condition**:

$$\lambda(g(x, y) - b) = 0$$

- ▶ Since we do not know if the constraint will be binding at the maximizer, we use the above equation to replace $\partial L / \partial \lambda = 0$

Theorem 18.3

Suppose that f and g are C^1 and (x^*, y^*) maximizes f on the set $g(x, y) \leq b$. If $g(x^*, y^*) = b$, further suppose that $\frac{\partial g}{\partial x}(x^*, y^*) \neq 0$ or $\frac{\partial g}{\partial y}(x^*, y^*) \neq 0$. In any case, form the Lagrangian function

$$L(x, y, \lambda) \equiv f(x, y) - \lambda(g(x, y) - b)$$

Then there is a multiplier λ^* such that

- ▶ (a) $\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) = 0$
- ▶ (b) $\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) = 0$
- ▶ (c) $\lambda^*[g(x^*, y^*) - b] = 0$
- ▶ (d) $\lambda^* \geq 0$
- ▶ (e) $g(x^*, y^*) \leq b$

Example 18.7: Consider the problem:

$$\begin{array}{ll}\max & f(x, y) = xy \\ \text{s.t.} & g(x, y) = x^2 + y^2 \leq 1\end{array}$$

The only critical point of g occurs at the origin – far away from the boundary of the constraint set $x^2 + y^2 = 1$. So the constraint qualification will be satisfied at any candidate for a solution. Form the Lagrangian function

$$L(x, y, \lambda) = xy - \lambda(x^2 + y^2 - 1)$$

and write out the first order conditions described in Theorem 18.3:

Example 18.7 :

$$\frac{\partial L}{\partial x} = y - 2\lambda x = 0$$

$$\frac{\partial L}{\partial y} = x - 2\lambda y = 0$$

$$\lambda(x^2 + y^2 - 1) = 0$$

$$x^2 + y^2 \leq 1$$

$$\lambda \geq 0$$

The first two equations yield

$$\lambda = \frac{y}{2x} = \frac{x}{2y} \quad \text{or} \quad x^2 = y^2$$

Example 18.7 :

- ▶ If $\lambda = 0$, then $x = y = 0$, which is a candidate for a solution.
- ▶ If $\lambda \neq 0$, then $x^2 + y^2 - 1 = 0 \Rightarrow x^2 = y^2 = \frac{1}{2}$, which gives $x = \pm \frac{1}{\sqrt{2}}, y = \pm \frac{1}{\sqrt{2}}$ then we find the following four candidates:

$$x = +\frac{1}{\sqrt{2}}, y = +\frac{1}{\sqrt{2}}, \lambda = +\frac{1}{2}$$

$$x = -\frac{1}{\sqrt{2}}, y = -\frac{1}{\sqrt{2}}, \lambda = +\frac{1}{2}$$

$$x = +\frac{1}{\sqrt{2}}, y = -\frac{1}{\sqrt{2}}, \lambda = -\frac{1}{2}$$

$$x = -\frac{1}{\sqrt{2}}, y = +\frac{1}{\sqrt{2}}, \lambda = -\frac{1}{2}$$

Example 18.7 : We disregard the last two candidates since they involve a negative multiplier. Plugging the three candidates into the object function, we find that

$$x = \frac{1}{\sqrt{2}}, y = \frac{1}{\sqrt{2}} \text{ and } x = -\frac{1}{\sqrt{2}}, y = -\frac{1}{\sqrt{2}}$$

are the solutions of our original problem. The two points with the negative multipliers are the solutions of the problem of minimizing xy on the constraint set $x^2 + y^2 \leq 1$.

► Problem setup:

$$\begin{array}{ll}\max & f(x) \\ \text{s.t.} & g_1(x) \leq b_1 \\ & g_2(x) \leq b_2 \\ & \vdots \\ & g_k(x) \leq b_k\end{array}$$

Theorem 18.4

Suppose x^* is a local maximizer of f on the constraint set.

- ▶ Assume the first k_0 constraints are binding at x^* and the last $k - k_0$ are not binding.
- ▶ Suppose the following NDCQ condition is satisfied at x^* : the rank of

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1}(x^*) & \cdots & \frac{\partial g_1}{\partial x_n}(x^*) \\ \vdots & \vdots & \vdots \\ \frac{\partial g_{k_0}}{\partial x_1}(x^*) & \cdots & \frac{\partial g_{k_0}}{\partial x_n}(x^*) \end{pmatrix}$$

is k_0 – as large as it can be.

Theorem 18.4 (Cond.)

Form the Lagrangian function:

$$L(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k) \equiv f(x) - \lambda_1[g_1(x) - b_1] - \dots - \lambda_k[g_k(x) - b_k].$$

Then there exist multipliers $\lambda_1^*, \dots, \lambda_k^*$ such that

- ▶ (a) $\frac{\partial L}{\partial x_1}(x^*, \lambda^*) = 0, \dots, \frac{\partial L}{\partial x_n}(x^*, \lambda^*) = 0$
- ▶ (b) $\lambda_1^*[g_1(x^*) - b_1] = 0, \dots, \lambda_k^*[g_k(x^*) - b_k] = 0$
- ▶ (c) $\lambda_1^* \geq 0, \dots, \lambda_k^* \geq 0$
- ▶ (d) $g_1(x^*) \leq b_1, \dots, g_k(x^*) \leq b_k$

- 1 Definition
- 2 Equality Constraints
- 3 Inequality Constraints
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- ▶ Some maximization problems involve both equality and inequality constraints
- ▶ Problem setup:

$$\begin{aligned} \max \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_1(\mathbf{x}) \leq b_1, \dots, g_k(\mathbf{x}) \leq b_k \\ & h_1(\mathbf{x}) = c_1, \dots, h_m(\mathbf{x}) = c_m \end{aligned}$$

Theorem 18.5

Suppose that $f, g_1, \dots, g_k, h_1, \dots, h_m$ are C^1 . Suppose that $x^* \in \mathbb{R}^n$ is a local maximizer of f on the constraint set.

- Assume the first k_0 constraints are binding at x^* and the last $k - k_0$ are not binding. Suppose the following NDCQ condition is satisfied at x^* : the rank of

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1}(x^*) & \cdots & \frac{\partial g_1}{\partial x_n}(x^*) \\ \vdots & \vdots & \vdots \\ \frac{\partial g_{k_0}}{\partial x_1}(x^*) & \cdots & \frac{\partial g_{k_0}}{\partial x_n}(x^*) \\ \frac{\partial h_1}{\partial x_1}(x^*) & \cdots & \frac{\partial h_1}{\partial x_n}(x^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1}(x^*) & \cdots & \frac{\partial h_m}{\partial x_n}(x^*) \end{pmatrix}$$

is $k_0 + m$ – as large as it can be.

Theorem 18.5 (Cond.)

Form the Lagrangian Function

$$L(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k, \mu_1, \dots, \mu_m) \equiv f(x) - \lambda_1[g_1(x) - b_1] - \dots - \lambda_k[g_k(x) - b_k] \\ - \mu_1[h_1(x) - c_1] - \dots - \mu_m[h_m(x) - c_m]$$

Then there exist multipliers $\lambda_1^*, \dots, \lambda_k^*, \mu_1^*, \dots, \mu_m^*$ such that:

- ▶ (a) $\frac{\partial L}{\partial x_1}(x^*, \lambda^*, \mu^*) = 0, \dots, \frac{\partial L}{\partial x_n}(x^*, \lambda^*, \mu^*) = 0$
- ▶ (b) $\lambda_1^*[g_1(x^*) - b_1] = 0, \dots, \lambda_k^*[g_k(x^*) - b_k] = 0$
- ▶ (c) $h_1(x^*) = c_1, \dots, h_m(x^*) = c_m$
- ▶ (d) $\lambda_1^* \geq 0, \dots, \lambda_k^* \geq 0$
- ▶ (e) $g_1(x^*) \leq b_1, \dots, g_k(x^*) \leq b_k$